

The Heat Is On!

Fall 2023 ARML Power Contest

The Solutions

1. $H(n, p, n)$ is the number of heats needed to determine the fastest p racers out of a total of n racers, when all n racers are allowed to race at one time! So race all the racers and determine which are the p fastest.
2. If you know which are the $n - 1$ fastest racers, there is only one racer left over who must be the slowest, so determining the $n - 1$ fastest racers also determines the n -th fastest.
3. In determining $H(n, n, n - 1)$, at least one racer will be left out of each race. So after one race there is at least one racer whose speed relative to any of the others is still unknown. So one heat will not be sufficient. If any racer left out of the first heat is also left out of the second, that racer's relative speed is still unknown and a third heat (at least) will be necessary. On the other hand, if any racer(s) left out of the first heat races in the second, and finishes in the exact same place(s) as any racers who were left out of the second heat originally placed in the first heat, their speed relative to the new racers is still unknown, also requiring at least a third heat. So in all possibilities, at least three heats are needed.

But three heats are always enough. In the first heat, race any $n - 1$ racers. Then in the second, replace the first-place winner of the first heat with the racer who hasn't raced. If that racer finishes in any place other than winning the second heat, the relative rankings of all n racers is now known. If the new racer *wins* the second heat, then the two heat winners must be the two fastest runners, because each has now beaten every other, and those others have all raced together so they can be put in their proper order by finish. Now race the two heat winners (and any other racers you wish) in the third heat, to determine who is the fastest and who is the second-fastest racer.

4. Call a racer "eliminated" if they race but do not win a heat. Since the goal is to determine the single fastest racer, any racer who loses a heat cannot be the fastest, so they truly are eliminated from consideration of being the fastest.

Next, note that a single heat can only eliminate $r - 1$ racers. Since r racers race in each heat, there are $r - 1$ who will not finish first in that heat, so at most $r - 1$ racers can be eliminated (it might be fewer, if some racers in that heat had already been eliminated previously, but were racing again anyway, "just for fun." The winner of a heat cannot be eliminated by that heat—if they are still in contention for first place overall, beating a bunch of other racers won't change that fact, and if they have been eliminated previously then this heat does not cause their elimination!

Now note that to determine the fastest racer, the other $n - 1$ racers must be eliminated. Since this can be done at most $r - 1$ contestants at-a-time, it must take at least $\lceil \frac{n-1}{r-1} \rceil$ heats to do the necessary eliminations. But a simple strategy will allow the fastest runner to be found in this many heats: the king-of-the-hill strategy. Race any r of the

racers, and the winner goes on to race $r - 1$ new racers, and so on, the winner of the current heat moving on to meet more challenges until there are no more racers who haven't raced. Since $r - 1$ new racers are eliminated each time, it takes $\lceil \frac{n-1}{r-1} \rceil$ heats until all but one are eliminated, and that one is declared to be the fastest racer.

5. First, we show that at least four heats will be needed to determine the order of the racers, and then a strategy will be given that never needs more than four heats to rank the racers.

In finding the minimum number of heats necessary to determine a given number of places, it is never necessary to race all the same racers in the same heat as had raced together in a previous heat. That is, there is never any need to repeat an entire heat, as no new information will be discovered about these racers and their relative speeds to each other or any other racers. So after three racers race in the first heat, the second heat will either include one or both of the two racers who haven't raced yet.

If only one new racer is included in the second heat, it could happen that after the second heat the relative speeds of all four of those racers will be known. For either the second heat includes the winner of the first heat, and the new racer might beat the previous winner, or the second heat must include the slowest racer in the first heat, and the new racer might prove to be even slower. So after two heats, it is possible that the relative speeds of four racers are known. But then since the fifth racer's speed hasn't been compared to any of those four, there is no way to compare to all four in one more heat. So it is impossible to determine the relative speeds of all five racers in three heats if there is only one new racer in the second heat.

On the other hand, if there are two new racers in the second heat, and just one holdover from the first heat, it is possible that the two new racers place "the same" relative to the holdover as the original racers did. That is, if the holdover from the first heat was the winner of that heat, and that racer also wins the second heat; or if the holdover was the middle racer in the first heat, that racer might again be the middle racer in the second heat. Call the faster of the two racers who only raced in the first heat $A1$ and the slower $A2$. Similarly, call the faster of the two racers who only raced in the second heat $B1$ and the slower $B2$. It is impossible after two heats to know the relative speeds of $A1$ and $B1$, because they have not raced each other, and either both beat or both lost to the racer who did race twice, so there is no chain of racers showing which of $A1$ or $B1$ is faster. The same is true about $A2$ and $B2$. But in only one more heat, with only three racers, it is not possible to resolve both the $A1$ - $B1$ and $A2$ - $B2$ questions.

So at least four heats are necessary.

Many strategies will allow the racers to be ranked that require at most four heats. One possibility is to race any three racers in the first heat, and call the racers $A1$ (fastest in the first heat), $A2$, and $A3$ (slowest in the first heat). Now race $A1$ against the other two racers, calling the faster of those two $B1$ and the slower $B2$. Then:

- If both $B1$ and $B2$ beat $A1$, the entire ranking is known: $B1$ (fastest), $B2$, $A1$, $A2$, $A3$ (slowest).

- If $B1$ beats $A1$ but $A1$ beats $B2$, then it is known that $B1$ is fastest and $A1$ is second-fastest; race $B2$, $A2$, and $A3$ to determine the complete rankings in three heats.
- if $A1$ beats $B1$, then in the third heat race $B1$, $B2$, and $A2$
 - If $A2$ is slowest, the ranking is known: $A1$ (fastest), $B1$, $B2$, $A2$, $A3$.
 - If $A2$ finishes in the middle, then the three fastest racers are $A1$, $B1$, $A2$, and the fourth heat will be between $B2$ and $A3$ and any other racer to determine fourth- and fifth- places.
 - If $A2$ wins the third heat, then the two fastest are $A1$ and $A2$, so race the remaining three in the fourth heat to determine their order.

Thus, in all cases, it is possible to determine the order of all five racers in at most four heats.

6. Similar to the explanation of problem 3, $H(n, n, n - 2) > 2$. There are two racers who aren't involved in the first heat, so after one heat it is impossible to know how they place among the other racers. In the second heat, the racers who didn't race in the first heat might place in the exact same places as those who raced in the first heat but not the second, in which case there is no way to compare the racers who only raced in one of the two heats to each other with only two heats.

On the other hand, there is always a way to place all the racers in three heats. Race $n - 2$ racers in the first heat, then replace two of them with the two who didn't compete in the first heat and race again. There are four racers who only competed in one heat. Since n is at least 6, $n - 2 \geq 4$, so those four racers can race in a third heat to determine their relative placing if it hasn't already been determined.

7. Following the reasoning of the proof of the previous problem, if $n - k$ racers are in each heat, then there are $2k$ racers who are only involved in one of the first two heats. They all have to be compared to each other, but once they are then the relation of each racer to each other racer will be known. So the number of racers in a heat must be at least large enough so that these $2k$ racers can be compared in one heat: $n - k \geq 2k$. Thus $n \geq 3k$ so setting $\boxed{M = 3k}$ will guarantee that n is large enough.
8. At least four heats are required. In fact, $H(6, 2, 3) \geq 4$, and $H(6, 3, 3)$ is certainly at least as large, since if the first 3 places are known, then the first two places are certainly known! To explain why $H(6, 2, 3) \geq 4$ label the winner and second-place finisher in the first heat as $A1$ and $A2$ respectively. The third-place finisher cannot be one of the top two places, and need no longer be considered. Consider the possibilities for the second heat:
 - If only one new racer is raced against both $A1$ and $A2$, then after this heat there are still two racers who have not raced yet, and no way to compare both of them with both $A1$ and $A2$ in one heat, so at least four heats are needed.
 - If two new racers compete against $A1$ and both lose (call them $B2$ and $B3$, in order of finish), then one of them ($B3$) is eliminated from consideration, and there

are now three candidates for second-place: A_2 , B_2 , and the competitor who has not yet raced, and all three must be compared. But if the last competitor is the fastest of these three, it is still unknown how they compare to A_1 . So again, at least four heats are needed.

- If the second heat consists of all three other racers, who finish in the order B_1 , B_2 , and B_3 , then all of A_1 , A_2 , B_1 , and B_2 still cannot be ranked in a single heat, so a fourth heat will be needed in this case also.

The casework above shows that at least four heats are required to find $H(6, 3, 3)$. Here is a scheme that finds the three fastest racers in only four heats. Choose any three racers for the first heat, and call them A_1 , A_2 , and A_3 from fastest to slowest. In the second heat, race A_2 against two new competitors. Of those two, let B_1 be the faster and B_2 the slower. Let C be the racer who has not raced yet.

- If the second heat finishes in the order B_1 , B_2 , A_2 , then there are at least three racers faster than A_2 , so A_2 is not among the fastest three, and certainly neither is A_3 . For the third heat race A_1 , B_1 , and C .
 - If this third heat finishes in the order A_1 , C , B_1 , then in only three heats we have found the order of the *four* fastest racers: A_1 , C , B_1 , B_2 , in that order. Similarly, if the third heat finishes in the order C , A_1 , B_1 , the top four places have been found.
 - If the third heat finishes in the order A_1 , B_1 , C , then A_1 is fastest, and B_1 is in second place. Race C , B_2 , and any third racer in the fourth heat to determine third- and fourth-place finishers. Similarly if the third heat ends in the order C , B_1 , A_1 .
 - If the third heat finishes in the order B_1 , A_1 , C , or the order B_1 , C , A_1 , then B_1 is fastest, and the fourth heat features A_1 , B_2 , and C to determine second, third, and fourth places.
- The second heat might finish in the order B_1 , A_2 , B_2 . Then A_1 , B_1 , and A_2 are each faster than both A_3 and B_2 , so those two are eliminated from further competition. In the third heat, race C against A_1 and B_1 . This will determine the top three places unless C is the slowest in this race. In that case, use the fourth heat to race C against A_2 (and any other racer) to determine who is in third place.
- Finally, A_2 might win the second heat. This eliminates B_2 and guarantees that either A_1 or C is the fastest racer. In the third heat, race A_3 , B_1 and C .
 - If C does not win this heat, then A_1 is fastest, A_2 is in second place, and the winner of the third heat is in third place.
 - If C does win the third heat, there is still a chance that C is one of the two fastest racers. So race A_1 , A_2 , and C to determine first, second, and third places.

9. Using the king-of-the-hill strategy, after $\lceil \frac{n-1}{r-1} \rceil$ heats, the fastest racer will be determined (compare with problem 4). There are also that many racers who placed second

in one of the heats, and the fastest of all those second-place finishers must be the second-fastest racer. Using the king-of-the-hill strategy with these $\lceil \frac{n-1}{r-1} \rceil$ racers will require another $\left\lceil \frac{\lceil \frac{n-1}{r-1} \rceil}{r-1} \right\rceil$ heats. So the two fastest racers can be found using at most $\left\lceil \frac{n-1}{r-1} \right\rceil + \left\lceil \frac{\lceil \frac{n-1}{r-1} \rceil}{r-1} \right\rceil$ heats.

It is a standard fact about the ceiling function that $\left\lceil \frac{\lceil a/b \rceil}{c} \right\rceil = \left\lceil \frac{a}{bc} \right\rceil$ for positive integers a , b , and c . Applying this to the formula derived above with $a = n - 1$ and $b = c = (r - 1)$ leads to the formula requested in the problem. Though it is not necessary for full credit, a proof of this standard fact is given here:

Given integers positive a , b , and c , choose integer k so that $k - 1$ is the quotient on dividing a by bc . That is,

$$(k - 1)bc < a \leq kbc.$$

By definition, $k - 1 < \frac{a}{bc} \leq k$ so $\lceil \frac{a}{bc} \rceil = k$. On the other hand, dividing by just b gives

$$(k - 1)c < \frac{a}{b} \leq kc$$

and since kc is an integer this shows that

$$(k - 1)c < \left\lceil \frac{a}{b} \right\rceil \leq kc.$$

Dividing this inequality by c and applying the ceiling function shows that $\left\lceil \frac{\lceil a/b \rceil}{c} \right\rceil = k$ as well.

10. If only two heats take place, then the top p racers must race in both heats. For contradiction, let $k < p$ and assume the racers who placed in the first k places in the first heat race again in the second heat, but the racer who was in place $k + 1$ did not race again. Then it is possible that the second heat finishes with the same top k racers, but a racer who did not take part in the first heat finishes in place $k + 1$. Then there is no way to compare whether the $(k + 1)$ -place finisher from the first heat or from the second heat should actually be in place $k + 1$ overall, and since $k + 1 \leq p$ the top p places have not been determined in only two heats. On the other hand, if the top p places from the first heat do race again in the second heat, then the second heat will determine the top p places, assuming all racers have raced at least once.

Put together, these statements determine the relationship between n , p , and r that make $H(n, p, r) = 2$. There are p racers who must be in both heats, and thus $r - p$ additional racers in each heat, and this must account for all racers. So $n \leq p + 2(r - p)$ or $\boxed{n \leq 2r - p}$.

11. To determine the two fastest racers from a group of $n = a + b$, one strategy would be to split the racers into a group of a and a group of b racers. The two fastest in each group

can now be found in $H(a, 2, 2)$ heats for the group of a , and $H(b, 2, 2)$ for the group of b racers. Now race the fastest of each group against each other. The winner will be the fastest racer overall, and the slower of those two can be raced against the second-place racer in the winner's group to determine who is the second-fastest racer overall. Thus, by using this divide-and-conquer strategy, at most $H(a, 2, 2) + H(b, 2, 2) + 2$ heats need to be run.

12. Several different approaches could work; here is an induction proof. The induction goes from n to $n + 2$ so requires two base cases:

- For $n = 2$, $H(2, 2, 2) = 1$ by problem 1, while $\lceil 3 \cdot 2/2 \rceil - 2 = 1$ also.
- For $n = 3$, $H(3, 2, 2) = H(3, 3, 2)$ by problem 2, which is found to equal 3 in problem 3. On the other hand, $\lceil 3 \cdot 3/2 \rceil - 2 = 5 - 2 = 3$, so the formula works in this case also.

(The formula works for $n = 1$ also, since no races are needed to determine who is fastest when there is just one racer!) Now assume the upper bound is correct for some n . Then

$$\begin{aligned}
 H(n + 2, 2, 2) &\leq H(n, 2, 2) + H(2, 2, 2) + 2 && \text{(by the previous problem)} \\
 &= H(n, 2, 2) + 1 + 2 && \text{(by problem 1)} \\
 &\leq \lceil 3n/2 \rceil - 2 + 3 && \text{(by the inductive assumption)} \\
 &= \lceil 3n/2 + 3 \rceil - 2 && \text{(since 3 is an integer)} \\
 &= \lceil 3(n + 2)/2 \rceil - 2
 \end{aligned}$$

which is the desired formula.

13. The answer is $\lceil \lg(n) \rceil$. Since each racer competes at most once in each round (and might not compete if they get a bye in that round), the maximum number of heats that a racer must win in order to be deemed the fastest is the same as the number of rounds. If there are at most 2^k racers to start, then there are at most 2^{k-1} racers who make it into the second round, at most 2^{k-2} racers in the third round, and so on. After $k - 1$ rounds have been completed, there are at most two racers remaining, and one last race determines the fastest. So if there are n racers, and n is at most 2^k , the winner need not win more than k heats. So $n \leq 2^k$ implies that the number of heats the eventual winner must win is at most $k = \lceil \lg(n) \rceil$.

Note that the total number of heats that need to be run, $H(n, 1, 2)$ is $n - 1$ as determined by problem 4. But the winner need not participate in all of them!

14. Assume a tournament has been arranged so that no matter who wins, they are guaranteed to have to win no more than k heats, and k is as small as possible. If $k = 1$ then there can only be two competitors, who race against each other to see who wins. If there is a third competitor (or more) then neither of the two competitors in the first heat can race again so there is no way to find out if any of the remaining competitors are faster than the winner of the first heat, thus the fastest racer cannot be

determined. So for $k = 1$ there are two competitors who race each other, which is a single-elimination bracket tournament.

Now for $k > 1$ consider the two competitors in the heat that determines the championship. This heat must be, at most, the k -th heat for both of them. For if one of the competitors will be racing for the $(k + 1)$ -st time (or later) and they win, they would not have been able to be declared fastest in fewer than $k + 1$ heats. Thus, each of the two competitors in the final championship heat will have competed in at most $k - 1$ previous heats.

This line of reasoning can be repeated inductively. Thus, there are at most four competitors who compete in $k - 2$ or more heats, eight that compete in $k - 3$ or more heats, and so forth. The only way to achieve these numbers is to eliminate half of the contestants at each stage, which is the set up of a single-elimination bracket tournament. Thus, no other race schedule allows the eventual winner to be guaranteed to run fewer races than the single-elimination bracket.

15. Note that the second-place racer must have lost to the fastest racer at some point. If they had lost to anyone other than the fastest racer, then there are at least two faster racers: the racer they lost to and the fastest racer. Now from problem 4 it takes $n - 1$ heats to determine the fastest racer, while from problems 13 and 14 it is known that there is no way for the fastest racer to be guaranteed to have won fewer than $\lceil \lg(n) \rceil$ races. Thus there may be as many as $\lceil \lg(n) \rceil$ racers who have only lost to the fastest racer. These racers can then race amongst themselves, taking $\lceil \lg(n) \rceil - 1$ heats to produce the second fastest racer. Thus, the fastest two racers can be guaranteed to be found in $n - 1 + \lceil \lg(n) \rceil - 1 = n - 2 + \lceil \lg(n) \rceil$ heats.
16. The first n racers can find the top p places in $H(n, p, r)$ heats. Now the newcomer wants to find out if they rate in the top p places, and if so, where in those top p places they belong.

Let $r^{k-1} \leq p < r^k$. We will exhibit a method of placing the newcomer within (or below!) the top p racers that requires k heats. Clearly if $k = 1$ then $p \leq r - 1$ so race the newcomer against all p of the fastest racers to determine their place (or that they are not among the top p if they lose to all of them). So if $k = 1$ the placement of the newcomer can be done in 1 heat.

Now proceed inductively. Assume that we can place the newcomer correctly within (or below!) the top p racers in at most k heats whenever $p < r^k$. For $r^k \leq p < r^{k+1}$, split the original p racers into groups of r^k racers. The fastest r^k racers go into the first group, the next fastest into the second group, and so on. If p is not a multiple of r^k then the group of slowest racers will have fewer than r^k racers in it, and since $p < r^{k+1}$ there are at most r^k groups. Race the newcomer against the slowest member of each group (except, if there are exactly r^k groups, leave out the slowest group). If the newcomer beats any of these racers, then they belong amongst the top $r^k - 1$ racers in the group of the fastest racer they beat, and their place within that group can now be determined (inductively) in at most k more heats (of course, everyone below them moves down one place).

If the newcomer loses to all racers in this heat, then they are either in the slowest group—which has at most $r^k - 1$ racers in it—or they are slower than all p of the fastest original racers. Since this slowest group has at most $r^k - 1$ racers, the newcomer's place in it can be determined in at most k more heats (and if they finish in last place among this group, then they are not in the fastest p places).

In any event it takes at most $k + 1$ heats to determine the placement of the newcomer when $p < r^{k+1}$, and the induction is complete. The proof has shown that if $r^{k-1} \leq p < r^k$ then at most k heats are required. Taking base- r logarithms, at most k heats are required when $k - 1 \leq \log_r(p) < k$ so $k - 1 = \lfloor \log_r(p) \rfloor$. Added to the original number of heats needed to find the fastest p of the original racers, the total is at most $H(n, p, r) + \lfloor \log_r(p) \rfloor + 1$ heats. (Note: this is a better bound than actually requested in the question; the requested result is correct, just not optimal!)

17. A very clever trick shows that $H(25, 3, 5) = 7$. First, it is known (problem 4) that it takes at least six heats to determine the fastest racer, and in any way of scheduling these six heats there are at least two racers who were second-place in their heat with the fastest racer, but have not themselves been compared. So $H(25, 2, 5) > 6$ and certainly $H(25, 3, 5) \geq H(25, 2, 5)$. So the answer can't be less than 7. But it turns out that seven heats is just barely enough! First, split the racers into five groups of five, and race each group in a heat. This creates five groups each of which is ordered within itself, say $A1$ through $A5$, $B1$ – $B5$, $\dots$, $E1$ – $E5$. Now race all the winners of these preliminary heats in the championship heat. Without loss of generality they finish in the order $A1$ (fastest), $B1$, $C1$, $D1$, $E1$. $A1$ is now proven to be the fastest racer. It is still possible for $A2$, $A3$, $B1$, $B2$, and $C1$ to be second and third—but every other racer has been proven to be slower than at least three others. So race these five in the consolation heat to determine second- and third-place.